



QUANTITATIVE METHODS 1 · CHAPTER 1

Data Presentation

Exercise series

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1. Statistical vocabulary

A survey asks 240 first-year students about programme, transport mode, satisfaction (1-5), number of absences and commuting time in minutes.

- 1 Identify the population, statistical unit and five variables.
- 2 Classify every variable as nominal, ordinal, discrete or continuous.
- 3 Give one possible value for each variable.

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

2. Population, sample and bias

KEDGE wants the average commuting time of all 1,200 first-year students. An online questionnaire is posted at 8 a.m.; 180 volunteers answer before noon.

- 1 Define the target population and sample.
- 2 Is the sample necessarily representative? Identify two plausible biases.
- 3 Propose a better sampling design.

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

3. Build a frequency distribution

The numbers of purchases made by 20 customers are: 0, 1, 2, 1, 3, 2, 2, 4, 1, 0, 2, 3, 1, 2, 5, 3, 2, 1, 4, 2.

- 1 Construct counts, relative frequencies, cumulative counts and cumulative frequencies.
- 2 Find the share with at most two purchases.
- 3 Choose an appropriate graph and justify it.

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

4. Unequal classes and responsible graphics

Delivery times are grouped as follows: $[0,5[: 20$; $[5,10[: 30$; $[10,20[: 40$; $[20,40[: 20$.

- 1 Calculate class widths, relative frequencies and frequency densities.
- 2 Explain why plotting counts as bar heights would be misleading.
- 3 Identify the modal class from densities.

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

5. Dashboard audit

A manager reports that satisfaction “doubled” because a truncated-axis chart rises from 3.8 to 4.2, and claims the sample represents the school although only members of one student association answered. List four statistical problems, redesign the communication and write a defensible conclusion.

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

6. Choosing the right chart

For each variable, choose a suitable graph and justify it: study programme; satisfaction from 1 to 5; monthly income; daily sales over 30 days; market shares of four brands.

Method prompt. Match the graph to the variable type and to the analytical question.

7. Cross-tabulation and conditional percentages

Among 120 students, 50 use public transport, including 30 women. Of the 70 other students, 35 are women.

- 1 Build the gender $\times$ transport table.
- 2 Compute the public-transport rate within each gender.
- 3 State whether an association appears present.

Method prompt. Conditional percentages use the total of the conditioning group as denominator.

8. Data-quality and communication protocol

A CRM extract contains duplicate customers, missing ages coded 999, inconsistent dates and revenue recorded in euros and thousands of euros. Propose a reproducible six-step protocol before producing a management dashboard.

Method prompt. Separate validation, cleaning, documentation and communication.

9. Continuous-assessment case: willingness to pay

Case study — 12 marks. A food retailer asks 200 customers for the maximum price they would pay for a sustainable lunch box.

Maximum price (€)	[20,25[	[25,30[	[30,40[	[40,50[	[50,60[	[60,80[
Customers	10	20	50	60	40	20

- 1 Identify the population, statistical unit, variable and variable type. (2 marks)
- 2 Calculate class widths, relative frequencies and frequency densities. (3 marks)
- 3 Explain how to construct the correct histogram and identify the modal class. (2 marks)
- 4 Calculate ascending and descending cumulative counts. Give the coordinates needed to draw both cumulative curves. (3 marks)
- 5 Find the highest price that at least 60% of surveyed customers would be willing to pay. Explain the percentile used. (2 marks)

Method prompt. Unequal-width classes require frequency density. For the pricing decision, translate “at least 60% can pay” into an ascending cumulative percentage.

10. Exam-style synthesis: reporting customer satisfaction

Independent case — 10 marks. Two stores collect satisfaction ratings on an ordered scale from 1 (very dissatisfied) to 5 (very satisfied).

Rating	1	2	3	4	5	Total
Store A	6	10	18	36	30	100
Store B	4	12	30	34	20	100

- 1 Identify the variable and its measurement scale. (1 mark)
- 2 Choose and justify one graph for comparing the two stores. Describe precisely how it should be constructed. (2 marks)
- 3 Compute, for each store, the percentages rated 4 or 5 and the cumulative percentage rated at most 2. (3 marks)
- 4 A manager says: "Store A is 50% better because it has 30 ratings of 5 versus 20." Criticise this claim. (2 marks)
- 5 Write a three-sentence statistical conclusion, including one limitation of the survey. (2 marks)

Method prompt. Keep the order of the categories, compare percentages because the store totals could differ in another dataset, and separate description from causation.